Principal Component Analysis (PCA)

PCA finds the most important directions in data and removes the rest.

Why Do We Need PCA?

Imagine dataset:

- Height
- Weight
- Age
- Salary
- Experience
- Education
- City
- 50 more columns 😵

👉 Problems:

- Slow training
- Hard to understand
- Some features are **duplicate (correlated)**

Solution (PCA)

👉 PCA:

- Removes unnecessary features
- Keeps only **important information**

Simple Real-Life Example

Imagine you take a photo 📷:

- Original = 4K image (very large size)
- Compressed = smaller size but still looks same

👉 PCA works like **data compression**

Core Idea

👉 Instead of using original features,
we create **new features (Principal Components)**

These are:

✔️ Combination of original features
✔️ Capture maximum information

Steps

1) Dataset

Feature	Example 1	Example 2	Example 3	Example 4
X_1	4	8	13	7
X_2	11	4	5	14

2) Find the Mean

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

Step 1: Calculate Mean

$$\bar{X}_1 = \frac{1}{4}(4+8+13+7) = 8,$$
$$\bar{X}_2 = \frac{1}{4}(11+4+5+14) = 8.5.$$

3) Find the covariance matrix

Covariance measures **relationship between two variables**:

- ➕ Positive → both increase together
- ➖ Negative → one increases, other decreases
- 0 → no relationship

👉 Example:

- Height ↑ and Weight ↑ → Positive covariance
- Temperature ↑ and Jacket sales ↓ → Negative covariance

- # What is a Covariance Matrix?
- When we have **multiple features**, we calculate covariance **for every pair**.
- 👉 Result = **Matrix (table of values)**

Principle Component Analysis – Solved Example

Step 2: Calculation of the covariance matrix.

$$S = \begin{bmatrix} \mathrm{Cov}(X_1, X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Cov}(X_2, X_2) \end{bmatrix}$$

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

$\overline{X_1} = 8$

$\overline{X_2} = 8.5$

$$\mathrm{Cov}(X_1, X_1) = \frac{1}{N-1}\sum_{k=1}^{N}(X_{1k} - \bar{X}_1)(X_{1k} - \bar{X}_1)$$
$$= \tfrac{1}{3}\left((4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2\right)$$
$$= 14$$

$\overline{X_2} = 8.5$

$$\mathrm{Cov}(X_1, X_2) = \frac{1}{N-1}\sum_{k=1}^{N}(X_{1k} - \bar{X}_1)(X_{2k} - \bar{X}_2)$$
$$= \tfrac{1}{3}((4-8)(11-8.5) + (8-8)(4-8.5)$$
$$+ (13-8)(5-8.5) + (7-8)(14-8.5)$$
$$= -11$$

$\overline{X_1} = 8$

$\overline{X_2} = 8.5$

$$\mathrm{Cov}(X_2, X_1) = \mathrm{Cov}(X_1, X_2)$$
$$= -11$$

$$\mathrm{Cov}(X_2, X_2) = \frac{1}{N-1}\sum_{k=1}^{N}(X_{2k} - \bar{X}_2)(X_{2k} - \bar{X}_2)$$
$$= \tfrac{1}{3}\left((11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2\right)$$
$$= 23$$

Step 2: Calculation of the covariance matrix.

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

$$\overline{X_1} = 8$$
$$\overline{X_2} = 8.5$$

$$S = \begin{bmatrix} \mathrm{Cov}(X_1, X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Cov}(X_2, X_2) \end{bmatrix}$$
$$= \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

4 Calculate Eigenvalues of the covariant matrix

Eigenvalues tell how much "important variation" exists in a direction

Step 3: Eigenvalues of the covariance matrix

The characteristic equation of the covariance matrix is,

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

$$\overline{X_1} = 8$$
$$\overline{X_2} = 8.5$$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

$$0 = \det(S - \lambda I)$$
$$= \begin{vmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{vmatrix}$$
$$= (14-\lambda)(23-\lambda) - (-11) \times (-11)$$
$$= \lambda^2 - 37\lambda + 201$$

✓ $\lambda_1 \approx ?$
✓ $\lambda_2 \approx ?$

4) Calculate Eigen Vectors
   Eigenvectors = directions where data naturally spreads

   Eigenvector = a direction that does NOT change when we apply a transformation (matrix)

   Only the **length changes**, not the direction.

Imagine:

- You stretch a rubber sheet
- Some directions stay straight (only longer/shorter)

👉 Those special directions = **Eigenvectors**

Step 4: Computation of the eigenvectors

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

$$U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \qquad \begin{bmatrix} 0 \\ 0 \end{bmatrix} = (S - \lambda\, I)\, U$$

$$= \begin{bmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$= \begin{bmatrix} (14-\lambda)u_1 - 11u_2 \\ -11u_1 + (23-\lambda)u_2 \end{bmatrix}$$

$$(14-\lambda)u_1 - 11u_2 = 0$$
$$-11u_1 + (23-\lambda)u_2 = 0$$

$$\overline{X_1} = 8$$
$$\overline{X_2} = 8.5$$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

$$\lambda_1 = 30.3849$$
$$\lambda_2 = 6.6151$$

Step 4: Computation of the eigenvectors

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

$$U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\frac{u_1}{11} = \frac{u_2}{14-\lambda} = t$$

$$u_1 = 11t, \quad u_2 = (14-\lambda)t$$

$$U = \begin{bmatrix} 11 \\ 14-\lambda \end{bmatrix}.$$

$$\overline{X_1} = 8$$
$$\overline{X_2} = 8.5$$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

$$\lambda_1 = 30.3849$$
$$\lambda_2 = 6.6151$$

5. Computation of the eigen vectors

Step 4: Computation of the eigenvectors

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

$$U_1 = \begin{bmatrix} 11 \\ 14 - \lambda_1 \end{bmatrix}.$$

- To find a unit eigenvector, we compute the length of U_1 which is given by,

$$\|U_1\| = \sqrt{11^2 + (14 - \lambda_1)^2}$$
$$= \sqrt{11^2 + (14 - 30.3849)^2}$$
$$= 19.7348$$

$$\overline{X_1} = 8$$

$$\overline{X_2} = 8.5$$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

$$\lambda_1 = 30.3849$$

$$\lambda_2 = 6.6151$$

$$e_1 = \begin{bmatrix} 11/\|U_1\| \\ (14 - \lambda_1)/\|U_1\| \end{bmatrix}$$

$$= \begin{bmatrix} 11/19.7348 \\ (14 - 30.3849)/19.7348 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}$$

Step 5.

Compute PCA

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Step 5: Computation of first principal components

F	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14

$$e_1^T \begin{bmatrix} X_{1k} - \bar{X}_1 \\ X_{2k} - \bar{X}_2 \end{bmatrix}$$

$$e_1 = \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}$$

$$\overline{X_1} = 8$$

$$\overline{X_2} = 8.5$$

$$e_1^T \begin{bmatrix} X_{1k} - \bar{X}_1 \\ X_{2k} - \bar{X}_2 \end{bmatrix} = \begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} X_{11} - \bar{X}_1 \\ X_{21} - \bar{X}_2 \end{bmatrix}$$

$$= 0.5574(X_{11} - \bar{X}_1) - 0.8303(X_{21} - \bar{X}_2)$$

$$= 0.5574(4 - 8) - 0.8303(11 - 8,5)$$

$$= -4.30535$$

$$e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}$$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

$$\lambda_1 = 30.3849$$

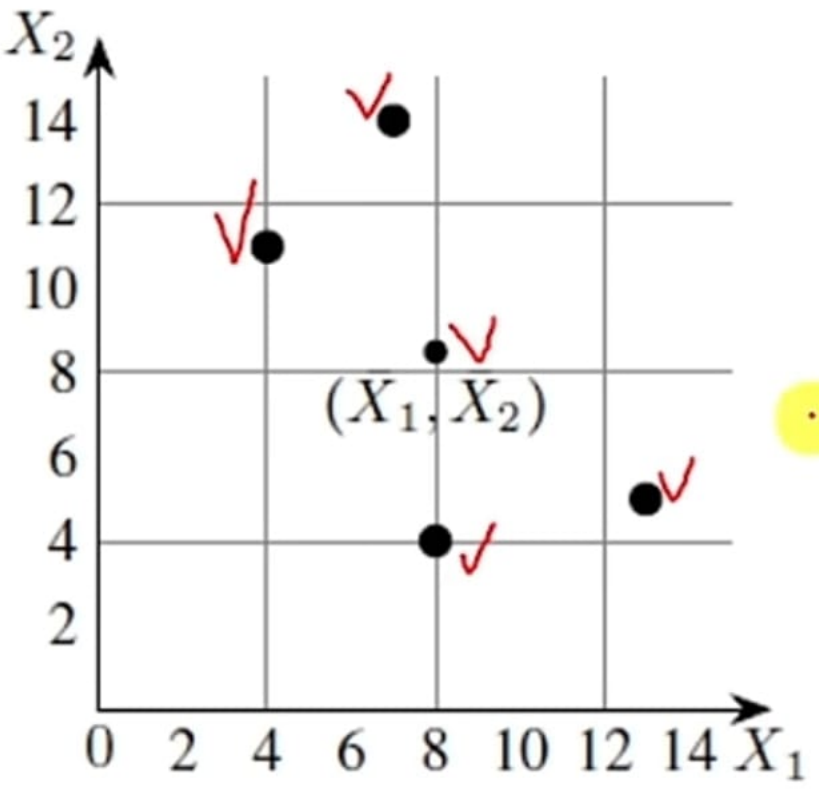
$$\lambda_2 = 6.6151$$

Feature	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14
First Principle Components	-4.3052	3.7361	5.6928	-5.1238

$$\overline{X_1} = 8$$

$$\overline{X_2} = 8.5$$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$



import numpy as np

```
# ----------------------------
# Step 1: Input Dataset
# ----------------------------
# X1 = [4, 8, 13, 7]
# X2 = [11, 4, 5, 14]

X = np.array([
   [4, 11],
   [8, 4],
   [13, 5],
   [7, 14]
], dtype=float)

print("Step 1: Original Data")
print(X)


# ----------------------------
# Step 2: Mean Centering
# ----------------------------
mean = np.mean(X, axis=0)
X_centered = X - mean

print("\nStep 2: Mean of each feature")
print(mean)

print("\nCentered Data (X - mean)")
print(X_centered)


# ----------------------------
# Step 3: Covariance Matrix
# ----------------------------
cov_matrix = np.cov(X_centered, rowvar=False)

print("\nStep 3: Covariance Matrix")
print(cov_matrix)


# ----------------------------
```

```
# Step 4: Eigenvalues & Eigenvectors
# -----------------------------
eigen_values, eigen_vectors = np.linalg.eig(cov_matrix)

print("\nStep 4: Eigenvalues")
print(eigen_values)

print("\nEigenvectors")
print(eigen_vectors)


# -----------------------------
# Step 5: Select Top Component
# -----------------------------
# Find index of largest eigenvalue
max_index = np.argmax(eigen_values)

# Select corresponding eigenvector
principal_component = eigen_vectors[:, max_index]

print("\nStep 5: Principal Component (Top Eigenvector)")
print(principal_component)


# -----------------------------
# Step 6: Project Data (Final PCA)
# -----------------------------
X_pca = np.dot(X_centered, principal_component)

print("\nStep 6: Final Reduced Data (1 Feature)")
print(X_pca)
```